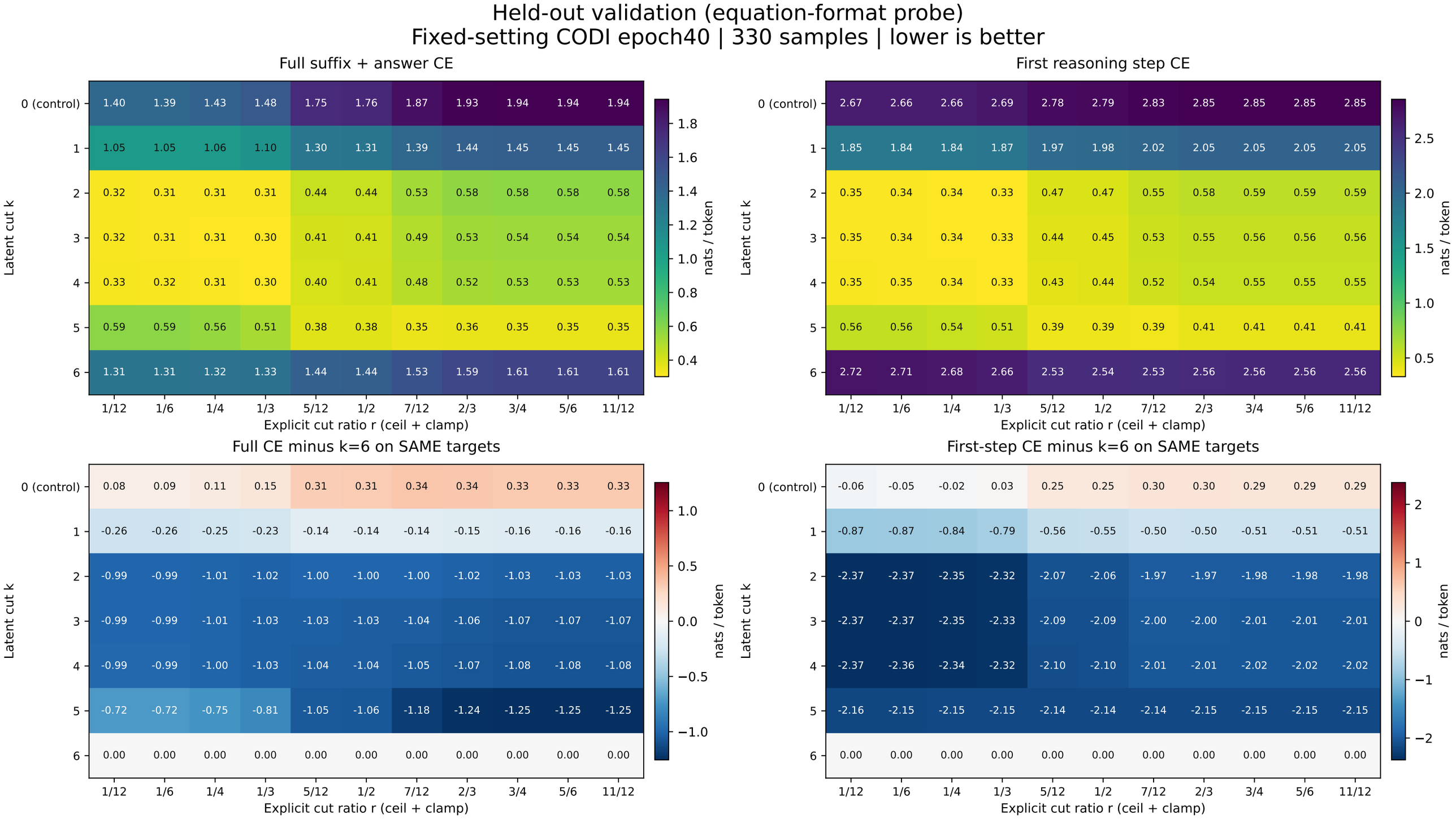
Held-out validation (equation-format probe)
Fixed-setting CODI epoch40 | 330 samples | lower is better
Full suffix + answer CE
0 (control) 1.40 1.39 1.43 1.48 1.75 1.76 1.87 1.93 1.94 1.94 1.94
1 1.05 1.05 1.06 1.10 1.30 1.31 1.39 1.44 1.45 1.45 1.45
2 0.32 0.31 0.31 0.31 0.44 0.44 0.53 0.58 0.58 0.58 0.58
3 0.32 0.31 0.31 0.30 0.41 0.41 0.49 0.53 0.54 0.54 0.54
4 0.33 0.32 0.31 0.30 0.40 0.41 0.48 0.52 0.53 0.53 0.53
5 0.59 0.59 0.56 0.51 0.38 0.38 0.35 0.36 0.35 0.35 0.35
6 1.31 1.31 1.32 1.33 1.44 1.44 1.53 1.59 1.61 1.61 1.61
Latent cut k
1/12 1/6 1/4 1/3 5/12 1/2 7/12 2/3 3/4 5/6 11/12
Explicit cut ratio r (ceil + clamp)
nats / token
1.8 1.6 1.4 1.2 1.0 0.8 0.6 0.4
First reasoning step CE
0 (control) 2.67 2.66 2.66 2.69 2.78 2.79 2.83 2.85 2.85 2.85 2.85
1 1.85 1.84 1.84 1.87 1.97 1.98 2.02 2.05 2.05 2.05 2.05
2 0.35 0.34 0.34 0.33 0.47 0.47 0.55 0.58 0.59 0.59 0.59
3 0.35 0.34 0.34 0.33 0.44 0.45 0.53 0.55 0.56 0.56 0.56
4 0.35 0.35 0.34 0.33 0.43 0.44 0.52 0.54 0.55 0.55 0.55
5 0.56 0.56 0.54 0.51 0.39 0.39 0.39 0.41 0.41 0.41 0.41
6 2.72 2.71 2.68 2.66 2.53 2.54 2.53 2.56 2.56 2.56 2.56
Latent cut k
1/12 1/6 1/4 1/3 5/12 1/2 7/12 2/3 3/4 5/6 11/12
Explicit cut ratio r (ceil + clamp)
nats / token
2.5 2.0 1.5 1.0 0.5
Full CE minus k=6 on SAME targets
0 (control) 0.08 0.09 0.11 0.15 0.31 0.31 0.34 0.34 0.33 0.33 0.33
1 -0.26 -0.26 -0.25 -0.23 -0.14 -0.14 -0.14 -0.15 -0.16 -0.16 -0.16
2 -0.99 -0.99 -1.01 -1.02 -1.00 -1.00 -1.00 -1.02 -1.03 -1.03 -1.03
3 -0.99 -0.99 -1.01 -1.03 -1.03 -1.03 -1.04 -1.06 -1.07 -1.07 -1.07
4 -0.99 -0.99 -1.00 -1.03 -1.04 -1.04 -1.05 -1.07 -1.08 -1.08 -1.08
5 -0.72 -0.72 -0.75 -0.81 -1.05 -1.06 -1.18 -1.24 -1.25 -1.25 -1.25
6 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Latent cut k
1/12 1/6 1/4 1/3 5/12 1/2 7/12 2/3 3/4 5/6 11/12
Explicit cut ratio r (ceil + clamp)
nats / token
1.0 0.5 0.0 −0.5 −1.0
First-step CE minus k=6 on SAME targets
0 (control) -0.06 -0.05 -0.02 0.03 0.25 0.25 0.30 0.30 0.29 0.29 0.29
1 -0.87 -0.87 -0.84 -0.79 -0.56 -0.55 -0.50 -0.50 -0.51 -0.51 -0.51
2 -2.37 -2.37 -2.35 -2.32 -2.07 -2.06 -1.97 -1.97 -1.98 -1.98 -1.98
3 -2.37 -2.37 -2.35 -2.33 -2.09 -2.09 -2.00 -2.00 -2.01 -2.01 -2.01
4 -2.37 -2.36 -2.34 -2.32 -2.10 -2.10 -2.01 -2.01 -2.02 -2.02 -2.02
5 -2.16 -2.15 -2.15 -2.15 -2.14 -2.14 -2.14 -2.15 -2.15 -2.15 -2.15
6 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Latent cut k
1/12 1/6 1/4 1/3 5/12 1/2 7/12 2/3 3/4 5/6 11/12
Explicit cut ratio r (ceil + clamp)
nats / token
2 1 0 −1 −2